

What is Google Dorking?

Google Dorking refers to using advanced search operators to find specific information that's not readily accessible through normal searches. These techniques leverage Google's extensive indexing capabilities to locate vulnerable systems, exposed data, or specific file types.

How Google Indexing Works

- Google's web crawlers continuously scan and index web content
- Over 130 trillion web pages indexed
- Processes over 3.5 billion searches daily
- Stores cached versions of web pages
- Indexes various file types (PDF, DOC, XLS, etc.)

Importance in Cybersecurity

- Vulnerability assessment
- Penetration testing
- Digital forensics
- Threat intelligence
- OSINT (Open Source Intelligence)

Basic Search Filters

Content-Based Search Operators

intext - Body Text Search

Searches for keywords within the main content of web pages.

Detailed Examples:

Basic usage

`intext:"username"`

`intext:"password"`

Multiple terms

`intext:"login" intext:"credentials"`

Combination with other operators

`intext:"admin" site:example.com`

allintext - Exact Phrase Requirement

Requires ALL specified keywords to appear in page content.

Detailed Examples:

```
# All terms must be present
allintext:"database connection string"

# Specific technical information
allintext:"API key" "authentication" "endpoint"

# Security-related searches
allintext:"vulnerability" "disclosure" "patch"
```

URL-Based Search Operators

inurl - URL Keyword Search

Finds specified keywords within the URL path.

Detailed Examples:

```
# Administrative interfaces
inurl:admin
inurl:login
inurl:dashboard
inurl:cp

# Configuration files
inurl:config
inurl:settings
inurl:configuration

# API endpoints
inurl:api
inurl:endpoint
inurl:rest
```

allinurl - Multiple URL Keywords

Requires ALL specified keywords to appear in the URL.

Detailed Examples:

```
# Specific path combinations
allinurl:admin login
allinurl:wp content uploads
allinurl:phpmyadmin config

# Database-related URLs
allinurl:php?id=
allinurl:asp?content=
```

Title-Based Search Operators

intitle - Page Title Search

Searches for keywords in HTML <title> tags.

Detailed Examples:

```
# Directory listings
intitle:"index of"
intitle:"directory listing"
```

```
# Login pages
intitle:"login"
intitle:"sign in"
intitle:"authentication"
```

```
# Error pages
intitle:"error"
intitle:"404"
intitle:"not found"
```

allintitle - Multiple Title Keywords

Requires ALL specified keywords in the page title.

Detailed Examples:

```
# Specific title combinations
allintitle:"server statistics"
allintitle:"web analytics"
allintitle:"admin panel"
```

Domain and Site Operators

site - Domain-Restricted Search

Limits searches to specific domains or TLDs.

Detailed Examples:

```
# Specific domain
site:github.com
site:example.com
```

```
# Top-Level Domains (TLDs)
site:.gov
site:.edu
site:.mil
```

```
# Subdomains
site:*.example.com
site:docs:*.com
```

```
# Multiple domains
site:com | site:org | site:net
```

File Type Operators

filetype - Specific File Extensions

Searches for specific file formats.

Complete File Type List:

```
# Documents
filetype:pdf
filetype:doc
filetype:docx
filetype:ppt
filetype:pptx
filetype:xls
filetype:xlsx
filetype:rtf
filetype:odt
```

```
# Data Files
filetype:sql
filetype:csv
filetype:json
filetype:xml
filetype:log
```

```
# Configuration Files
filetype:conf
filetype:config
filetype:ini
filetype:yml
filetype:yaml
filetype:env
```

```
# Source Code
filetype:php
filetype:js
filetype:py
filetype:java
filetype:c
filetype:cpp
```

```
# Archives
filetype:zip
filetype:rar
filetype:tar
filetype:gz
```

```
# Database Files
filetype:mdb
filetype:accdb
filetype:db
filetype:sqlite
```

Practical Examples:

```
# Confidential documents
filetype:pdf "confidential" "internal use only"
```

```
# Database dumps
filetype:sql "CREATE TABLE" "INSERT INTO"
```

```
# Configuration files with credentials
filetype:env "DB_PASSWORD" "API_KEY"

# Log files with sensitive data
filetype:log "password" "username"
```

Advanced Search Operators

Link and Relationship Operators

link - Backlink Discovery

Finds pages linking to a specific URL.

Detailed Examples:

```
# Basic backlink check
link:example.com
```

```
# Competitor analysis
link:competitor.com
```

```
# Authority checking
link:government.gov
```

related - Similar Sites

Finds websites similar to specified domain.

Detailed Examples:

```
# Find competitors
related:amazon.com
```

```
# Similar services
related:github.com
```

```
# Industry peers
related:microsoft.com
```

Numeric and Range Operators

numrange - Number Range Search

Searches for specific numerical values or ranges.

Detailed Examples:

```
# Price ranges
"product" numrange:100-500
"cost" numrange:1000-5000
```

```
# Version numbers
"version" numrange:2.0-3.0
"release" numrange:2020-2023
```

```
# Technical specifications
"memory" numrange:8-16
"storage" numrange:500-1000
```

daterange - Date Range Search

Searches within specific date ranges (Julian calendar format).

Detailed Examples:

```
# Julian date format
daterange:2459545-2459580
```

```
# Combined with filetype
filetype:pdf daterange:2459545-2459580
```

Date Filters

before / after - Date Range Filtering

Searches content within specific date ranges.

Detailed Examples:

```
# Basic date ranges
after:2023-01-01 before:2023-12-31
```

```
# Recent content
after:2024-01-01
```

```
# Historical content
before:2020-01-01
```

```
# Combined searches
filetype:pdf after:2023-01-01 "confidential"
```

Anchor Text Operators

inanchor / allinanchor - Link Text Search

Searches for text in hyperlinks pointing to pages.

Detailed Examples:

```
# Download links
inanchor:"download"
inanchor:"click here"
```

```
# Security-related anchors
allinanchor:"free download" "full version"
```

```
# Navigation elements
inanchor:"home" "about" "contact"
```

Cache and Info Operators

cache - **Cached Page View**

Shows Google's stored version of a page.

Detailed Examples:

```
# View cached version
```

```
cache:example.com
```

```
# Specific page cache
```

```
cache:example.com/admin
```

```
# Compare current vs cached
```

```
cache:example.com/login
```

info - **Page Information**

Shows information Google has about a page.

Detailed Examples:

```
# Basic info
```

```
info:example.com
```

```
# Multiple pages
```

```
info:example.com/page1 info:example.com/page2
```

Location-Based Operators

location - **Geographic Targeting**

Filters results by geographic location.

Detailed Examples:

```
# Country-specific
```

```
location:US
```

```
location:UK
```

```
location:DE
```

```
# Combined with other operators
```

```
site:.gov location:US "confidential"
```

source - **News Source Filtering**

Filters news results by specific sources.

Detailed Examples:

```
# Specific news sources
```

```
source:bbc
```

```
source:reuters
```

```
source:associated_press
```

Boolean Logic & Combinations

Basic Boolean Operators

OR Operator (|)

Broadens search to include multiple alternatives.

Detailed Examples:

```
# Multiple domains  
site:facebook.com | site:twitter.com | site:linkedin.com
```

```
# Multiple file types  
filetype:pdf | filetype:doc | filetype:docx
```

```
# Multiple keywords  
"username" | "user" | "login"
```

AND Operator (&)

Narrows search by requiring multiple conditions.

Detailed Examples:

```
# Multiple requirements  
site:github.com & intext:"API" & intext:"key"
```

```
# Complex combinations  
filetype:sql & intext:"password" & intext:"users"
```

NOT Operator (-)

Excludes specific terms from results.

Detailed Examples:

```
# Exclude specific sites  
"database" -site:github.com -site:stackoverflow.com
```

```
# Exclude file types  
"config" -filetype:pdf -filetype:doc
```

```
# Exclude terms  
"admin" -"test" -"demo"
```


Advanced Boolean Combinations

Parentheses for Grouping

Controls operator precedence and complex logic.

Detailed Examples:

```
# Complex domain searches
(site:facebook.com | site:twitter.com) & (intext:"login" | intext:"signin")

# Multiple conditions
(filetype:pdf | filetype:doc) & ("confidential" | "secret") & -"public"

# Nested logic
(site:.gov | site:.edu) & (intext:"research" | intext:"study") & -"publication"
```

Sequential Filtering

Combining multiple operators for precision.

Detailed Examples:

```
# Progressive filtering
site:.edu intext:"database" filetype:sql -"test"

# Multiple exclusion criteria
"password" filetype:txt -site:pastebin.com -site:github.com
```

Inclusion/Exclusion Techniques

Force Inclusion (+)

Requires specific terms to be present.

Detailed Examples:

```
# Required terms
+site:facebook.* +intext:"admin"

# Multiple requirements
+"API" +"key" +"authentication"
```

Strategic Exclusion

Removing unwanted results effectively.

Detailed Examples:

```
# Exclude common false positives
"index of" -"html" -"htm" -"php"
```

```
# Exclude specific content types
"password" -filetype:html -filetype:php
```

Practical Use Cases

Security Vulnerability Discovery

Exposed Configuration Files

```
# Common config files
inurl:"config.xml" | inurl:"config.json" | inurl:".env"
filetype:env DB_PASSWORD
inurl:"wp-config.php"
"database_password" filetype:cfg

# Framework-specific configs
"settings.py" "SECRET_KEY"
"config/database.yml"
"application.properties" "spring.datasource.password"
```

Open Directory Listings

```
# Basic directory listings
intitle:"index of"
"parent directory"
"directory listing for"

# Specific content types
intitle:"index of" "mp3"
intitle:"index of" "pdf"
intitle:"index of" "backup"

# Advanced directory searches
intitle:"index of" "site.tar.gz"
intitle:"index of" "database.sql"
intitle:"index of" "backup.zip"
```

Exposed Administrative Interfaces

```
# Common admin panels
inurl:/admin/
inurl:/phpmyadmin/
inurl:/cpanel/
inurl:/webmin/

# Device-specific interfaces
intitle:"router" inurl:"login"
intitle:"camera" inurl:"login"
"network device" "configuration" inurl:admin
```

Information Gathering

Employee and Organizational Data

```
# Employee directories
"employee directory" site:company.com
"staff list" filetype:xls site:org.com
"organizational chart" filetype:pdf
```

```
# Contact information
"@company.com" "phone" "extension"
"contact us" filetype:vcf
"email" "directory" site:edu
```

Financial and Business Information

```
# Financial documents
filetype:pdf "financial report" "Q1 2024"
"profit and loss" filetype:xlsx
"balance sheet" "confidential"

# Business plans
"business plan" "proprietary" "competitive analysis"
"strategic roadmap" "internal use only"
```

Technical Information Discovery

Network and Infrastructure Data

```
# Network diagrams
"network diagram" "infrastructure" filetype:vsd | filetype:pdf
"topology" "internal network"
"rack layout" "data center"

# Server information
"server status" "internal"
"monitoring" "dashboard" inurl:8080
```

API and Development Data

```
# API documentation
"API documentation" "endpoints" "authentication"
"Swagger" "OpenAPI" filetype:json | filetype:yaml

# Source code exposure
"source code" "repository" inurl:.git
filetype:sql "CREATE TABLE" "users"
```

Security Research Techniques

Credential Discovery

Password and Key Exposure

```
# Various credential formats
"password" filetype:txt | filetype:log | filetype:sql
"api_key" "="
"ssh-rsa" "PRIVATE KEY"
"BEGIN RSA PRIVATE KEY"

# Platform-specific keys
"AWS_ACCESS_KEY"
"AZURE_STORAGE_CONNECTION_STRING"
"GOOGLE_API_KEY"
```

Authentication Data

```
# Login credentials
"login" "password" filetype:csv
"username" "password" filetype:xlsx
"authentication" "credentials" filetype:xml

# Session data
"session_id" "cookie"
"token" "expires"
```

System Vulnerability Identification

Version Disclosure

```
# Software versions
"Powered by" "Version"
"Apache" "Server Version"
"nginx" "built on"

# Vulnerability-specific searches
"vulnerable version" "update required"
"security patch" "CVE-2024"
```

Error Messages and Debug Information

```
# Debug information
"debug mode" "true"
"stack trace" "exception"
"error occurred at"

# Database errors
"mysql_fetch_array()"
"ODBC Driver" "error"
"SQLException"
```

Network Service Discovery

Open Ports and Services

```
# Service banners
"Apache" "Server at" "port"
"nginx" "welcome to"
"ISS" "Windows Server"

# Protocol-specific
"ftp" "anonymous" "login"
"telnet" "connected to"
"ssh" "authorized"
```

Device-Specific Interfaces

```
# IoT devices
"router login" "default password"
"camera" "web interface"
"printer" "configuration page"

# Industrial control systems
"SCADA" "HMI" "login"
"PLC" "interface" "control"
```

Information Gathering Methods

Corporate Intelligence

Internal Documentation

```
# Policy and procedure documents
"internal policy" "confidential" filetype:pdf
"employee handbook" "proprietary"
"standard operating procedure" "SOP"

# Strategic documents
"strategic plan" "internal"
"roadmap" "product development"
"competitive analysis" "restricted"
```

Communication and Meeting Data

```
# Meeting minutes
"meeting minutes" "action items" filetype:doc
"board meeting" "confidential"

# Email and communication
"internal memo" "distribution list"
"announcement" "employees only"
```

Educational and Research Data

Academic Resources

Research papers

filetype:pdf "dissertation" "thesis"

"research paper" "unpublished"

"study" "preliminary findings"

Course materials

"syllabus" "lecture notes" filetype:pdf

"assignment" "solution manual"

Institutional Data

University information

"campus" "faculty" "directory"

"student records" "FERPA"

"research data" "dataset"